Jamieson Mansker

+1 **(910) 849-5693 | jamieson.mansker@wsu.edu**

Computer Science student specializing in embedded systems and full-stack web development, with hands-on experience building real system devices with ML data inference.

Education

Washington State University **Pullman, WA**

B.S. **Computer Science**, Minoring in **Electrical Engineering** and **Math** **Fall 2023 - Spring 2028**

3rd Year | Presidential Honors Award Fall 2023 | GPA 3.3

Relevant Coursework: Advanced Data Structures C/C++ | Object Oriented Software Princ. C# | Software Engineering I | Algorithm Design and Analysis | Systems Programming C/C++ | Computer Architecture MIPS

Personal Projects

github.com/JamiesonMan

Computer Vision Bird Nest Monitoring System - PyTorch, Scikit, REST API, OpenCV **Summer 2025**

- Designed and deployed a **fault-tolerant embedded Linux system** on a Raspberry Pi Zero that continuously captured and streamed live video on bootup, using **systemd services and timers** to ensure automatic recovery and high uptime.
- Created a **distributed edge-to-server pipeline** where the device scheduled frame captures and transmitted data to a remote server for **PyTorch-based inference on a CNN model** I created.
- Automated data collection for future models; final model produced **94% accurate** inferred results.
- Collected **75+ hours** of behavioral science data on Dark Eyed Junko bird species during nesting cycle.

Full Stack Portfolio Website (liamthechemist.com) – Flask, Docker, SQLite, WSGI **Summer 2025**

- Designed and developed a **Flask-based full-stack website** with a role-based admin interface, enabling an admin user to manage site content dynamically.
- Implemented a **RESTful backend** with persistent data storage using **SQLite**, focusing on data integrity, security, and maintainability.
- Maintained a **Docker Image**, ensuring a portable environment for development and local testing.
- Features embedded client Desmos projects to provide an **interactable and explorable environment**, and a subscriber based automated **email notification system** when new content is released and **MFA logins**.

Chess Engine – C++, Perft, Design Patterns, CMake **Summer 2025 – Present**

- Developed game of chess in a memory efficient architecture suitable for a Chess Engine**.**
- Implemented UCI communication protocol allowing Engine to be modularly linked to **dozens** of Chess GUIs.

Technical Skills

C, C++, Python**,** C#, Javascript, SQL, REST API Design, React, AWS EC2, AWS Lambda**,** Linux (Ubuntu), Docker, Github Actions, .NET, Firebase, Multithreading & Concurrency, SSH, CMake, Debugging, SQLite

Work Experience

Dairy Queen – Part Time Closing Team Member **Spring 2025 - Fall 2025**

Balanced 15-hour work weeks in a fast-paced environment while fully enrolled in course work. **Pullman WA**

Certificates and Awards

AWS Certified Cloud Practitioner **2025 – 2028**

Extracurriculars

Crimson Code Hackathon | Spring 2026

Crimson Robotics Club | Present

WSU Collegian Chess Team | Fall 2023 - Present

WSU Rowing Team | Spring 2024 – Spring 2025